

Himansh High-Altitude Glaciological Research Station

Annual Glacio-Hydrological Balance and Mass Budget Assessment · Chandra Basin

1.0 Cryospheric Setting and Hydrological Importance

Established in 2016 at an altitude of 4,080 meters (13,386 ft) in the Chandra River basin of the Spiti valley, Himansh ('a slice of ice') is India's dedicated high-altitude research station. It serves as an active hub for continuous monitoring of Himalayan benchmark glaciers including Sutri Dhaka, Batal, Samudra Tapu, and Gepang Gath.

The Western Himalayan glaciers serve as crucial water towers for the Indus river network. In-situ mass balance stakes, automated acoustic depth sounders, and ground penetrating radar (GPR) arrays provide critical ground validation for satellite gravimetry (GRACE-FO) and radar altimetry missions.

2.0 Benchmark Glacier Mass Balance & Snow Water Equivalent (SWE)

Glacier	Area (km²)	ELA (m a.s.l.)	Winter Bal (m w.e.)	Summer Bal (m w.e.)	Net Balance
Sutri Dhaka	19.8	5120	+1.14	-1.82	-0.68
Batal	4.2	5080	+0.92	-1.74	-0.82
Samudra Tapu	65.4	5240	+1.28	-1.96	-0.68
Gepang Gath	14.1	4980	+1.05	-1.62	-0.57

3.0 Hydrological Meltwater Discharge Telemetry

Automated ultrasonic stage-level sensors installed at the Sutri Dhaka proglacial stream logged peak discharge during late July (18.4 cubic meters per second), coinciding with highest seasonal air temperatures. Suspended sediment concentration reached 3.8 g/L during peak ablation periods.

These empirical observations provide essential baseline calibration for downstream hydropower installations and flood early warning models across the Himachal Pradesh mountain catchments.